

# APS UPGRADE - BEAMLINE ENGINEERING OVERVIEW\*

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## Abstract

The Advanced Photon Source Upgrade (APSU) has transformed the APS into a 4th generation light source. The new 6 GeV, 200 mA multi-bend achromat (MBA) storage ring, along with injector improvements and new front ends will provide an increased brightness and an orders-of-magnitude improvement in coherent flux over the current facility.

To take advantage of these new capabilities, we have designed and constructed nine new “feature beamlines” and implemented numerous additional beamline enhancements, all while ensuring the compatibility of existing programs. This paper will present a comprehensive overview of the APSU beamline scope, focusing on successes, challenges, and lessons learned.

## FEATURE BEAMLINES

Nine “feature beamlines” were chosen to showcase “best in class” performance, taking full advantage of the new accelerator. The feature beamline scope included:

- New enclosures and utilities
- New Photon delivery systems
- New Optics
- New Beamline instrumentation
- New Controls systems



Figure 1: New Long Beamline Building (LBB).

Two of these beamlines, the InSitu Nanoprobe (ISN) and the High Energy X-ray Microscope (HEXM), extend into the new Long Beamline Building (Fig. 1) to enable long-distance optics, low-background environments, and enhanced stability for nanoscale focusing and high-energy diffraction/imaging.

The Coherent High Energy X-ray (CHEX) beamline has four independently operating branches with eight experimental stations from two canted undulators. The tunable-energy branch (28-ID-A/B/C) operates from 15-60 keV with a horizontally deflecting DCM. Three selectable-energy branches (28-ID-A/D/E, 28-ID-A/F, and 28-ID-A/G), operate at fixed energies between 15 to 60 keV utilizing single crystal monochromators.

The remaining feature beamlines include POLAR, XPCS, CSSI, 3DMN, ATOMIC, and PtychoProbe, spanning coherent imaging, high-energy scattering, time-resolved dynamics, and operando studies. The overall project Key Performance Parameters (KPPs) required demonstration of:

- Brightness at a defined energy and verification of the usable energy range.
- ISN: a 50 nm × 50 nm focal spot at the sample.

All feature beamlines met or exceeded their KPPs!

## ENHANCEMENT BEAMLINES

Fifteen existing beamlines received major upgrades. The specific enhancements were chosen to get the best impact for money spent:

- Canted geometries to create independent branches, requiring new front-ends, shutters, primary slits, and branch enclosures; in most cases requiring significant redesign of their photon delivery system.
- New optics (monochromators, mirrors, multilayers, refractive lenses), upgraded absorbers and shielding.
- New/modernized instruments (diffractometers, detectors, sample environments).
- Controls and timing updates, beam-based alignment, and improved thermal-mechanical stability.

In total, approximately 60 instruments and about 30 optical subsystems were delivered or refurbished across the beamlines, leveraging reuse where appropriate.

## BENDING MAGNET (BM) BEAMLINES

Due to the shift in the bending magnet source in the new storage ring, all BM beamlines were required to move inboard by 42 mm with an additional 0.5 mrad inboard angular shift. This realignment forced the existing components closer to the storage ring shielding wall in the first optical enclosures necessitating rerouting of utilities. The additional angular shift had a larger impact on downstream transport pipe, wall openings, and transitional shielding as the beam center deviated further from the original trajectory.

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## DO-NO-HARM BEAMLINES

The APS-U committed to ensuring that all remaining beamlines would be compatible with the new APS-U source parameters. This undertaking affected 12 ID beamlines, and all 18 BM beamlines. The Do-No Harm scope was among the most complex tasks, given decades of design diversity and incomplete legacy records. Each beamline required:

- Full radiation safety shielding review against updated source parameters.
- Updated photon delivery system ray tracings for new source parameters
- Thermal analyses for higher power densities
- Aperture/clearance checks.
- Personnel safety systems (PSS) and egress, comprehensive interlock logic/hardware reviews and code compliance updates.
- Field verification of existing components, creation of missing 3D models, and controlled as-built drawings to close documentation gaps.
- Survey & Alignment

## DESIGN STANDARDIZATION

Standardization of components and processes was an absolute necessity when taking on a project this big. All beamlines are unique, but we identified many places where shared designs were possible. Standard documents, including reports, presentations and ray tracings (Fig. 2) streamlined reviews and overall communication.

Harvested general components acquired from the old accelerator and reconfigured beamlines (e.g. bellows, gate valves, ion pumps, vacuum spools and support assemblies) were also utilized. Inspected, cleaned, leak-checked, reconditioned and requalified for installation, having these components on hand saved enormous amounts of money and effort.

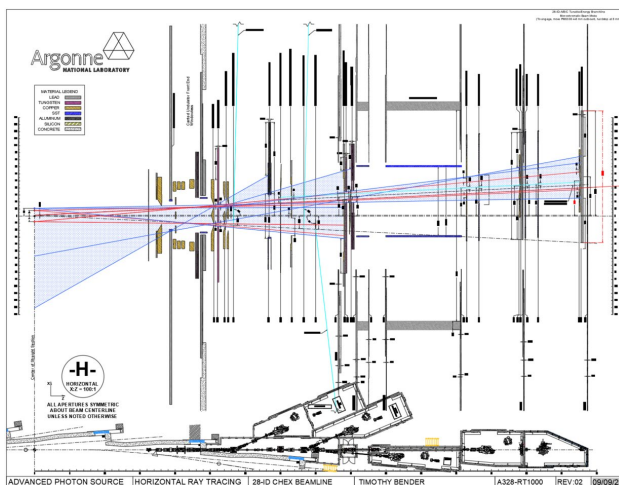


Figure 2: Typical beamline ray tracing

## COMMISSIONING AND READINESS

The transition plan from installation to user operations was followed for each individual beamline:

- Checkout of Photon Delivery Systems; Vacuum integrity, utilities/cooling, interlock/PSS validation, shielding checks, alignment of front-end apertures and primary optics.
- Beamline Readiness Review
- Shielding Verification; First light and conditioning, white-beam acceptance; thermal conditioning of mirrors and beam-based alignment and stabilization.
- Start Technical Commissioning (with beam); End-station/science commissioning; Monochromator energy and resolution checks; radiation surveys.
- Technical Commissioning (KPP measurements) (brightness, energy range, focal spot); radiation surveys.

70 ports have undergone independent safety and operational readiness reviews prior to restart.

## LESSONS LEARNED

There were many lessons learned but a few stand out:

- Dedicated teams should be established to focus on engineering, procurement, financial reporting and Q/A.
- Separate engineering from project management tasks!
- Dedicated storage, assembly and testing space throughout the project duration to avoid time-consuming relocations and misplaced items.
- Recognize that team members are unique and not interchangeable; assign support based on specific expertise, such as providing experienced personnel for tasks like drafting Statements of Work (SOWs).
- Standardize components, naming conventions, drawing numbers to streamline communication and organization.
- It's never too late to step back and reassess plan going forward.

## CONCLUSION

The APS-U delivered a fourth-generation source and a transformed beamline suite. Nine feature beamlines exceeded KPPs, 15 enhancement beamlines gained major capabilities, and all remaining beamlines were validated under the do-no-harm program. In total, 70 ports were qualified for safe operation. Standardization, rigorous documentation, strategic reuse and clear organizational structures underpinned on-schedule commissioning and provide a foundation for sustained operations [1].

## ACKNOWLEDGMENTS

We thank the APS-U accelerator, beamline, engineering, controls, safety and science teams, partner institutions, vendors and the user community for their contributions to this monumental achievement.

## REFERENCES

- [1] Advanced Photon Source Upgrade (APS-U), <https://www.aps.anl.gov/APS-Upgrade>